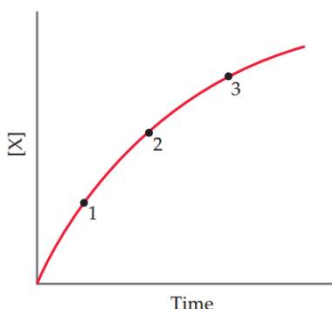


Chapter 14 Homework

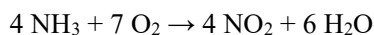
Part A multiple choice (select the one that is best in each case. 1 point/question)

- 1) Which three factors directly affect the rate of a chemical reaction?
- A) temperature, product concentration, and container volume
B) temperature, product concentration, and pressure
C) temperature, reactant concentration, and catalyst
D) temperature, pressure, and humidity
E) catalyst, product concentration, and container volume
- 2) Consider the following graph of the concentration of a substance X over time. Which of the following statements is true?



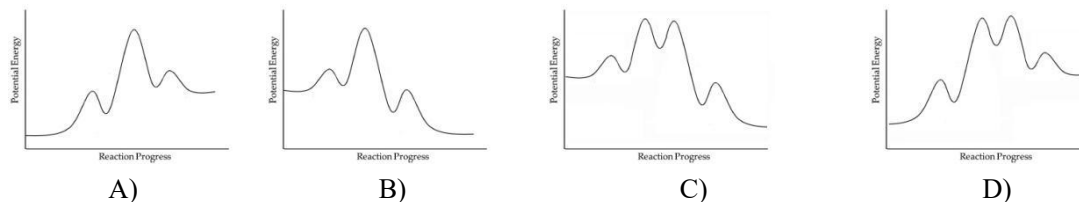
- A) The rate of the reaction remains the same as time progresses.
B) X is a product of the reaction.
C) As time progresses, the curve will eventually turn downward toward the x-axis.
D) The average rate between points 1 and 2 is smaller than the average rate between points 1 and 3.
- 3) At a certain time in a reaction, substance A is disappearing at a rate of $4.0 \times 10^{-2} M/s$, substance B is appearing at a rate of $2.0 \times 10^{-2} M/s$, and substance C is appearing at a rate of $6.0 \times 10^{-2} M/s$. Which of the following could be the stoichiometry for the reaction being studied?
- A) $2 A + B \rightarrow 3 C$ B) $A \rightarrow 2 B + 3 C$ C) $2 A \rightarrow B + 3 C$
D) $4 A \rightarrow 2 B + 3 C$ E) $A + 2 B \rightarrow 3 C$
- 4) Consider the combustion of $H_2(g)$: $2 H_2(g) + O_2(g) \rightarrow 2 H_2O(g)$
If hydrogen is burning at the rate of $4.6 \text{ mol} \cdot \text{s}^{-1}$, what are the rates of consumption of oxygen and formation of water vapor?
- A) $-2.3 \text{ mol} \cdot \text{s}^{-1} (O_2)$ and $+2.3 \text{ mol} \cdot \text{s}^{-1} (H_2O)$
B) $-4.6 \text{ mol} \cdot \text{s}^{-1} (O_2)$ and $-4.6 \text{ mol} \cdot \text{s}^{-1} (H_2O)$
C) $+2.3 \text{ mol} \cdot \text{s}^{-1} (O_2)$ and $+4.6 \text{ mol} \cdot \text{s}^{-1} (H_2O)$
D) $-4.6 \text{ mol} \cdot \text{s}^{-1} (O_2)$ and $+4.6 \text{ mol} \cdot \text{s}^{-1} (H_2O)$
E) $-2.3 \text{ mol} \cdot \text{s}^{-1} (O_2)$ and $-4.6 \text{ mol} \cdot \text{s}^{-1} (H_2O)$
- 5) What variables does the *rate constant* depend on?
- I. concentrations of the products
II. concentrations of the reactants
III. reaction temperature
IV. magnitude of E_a
- A) I B) II C) III D) IV E) Both III and IV
- 6) Which one of the following is not a valid expression for the rate of the reaction below?

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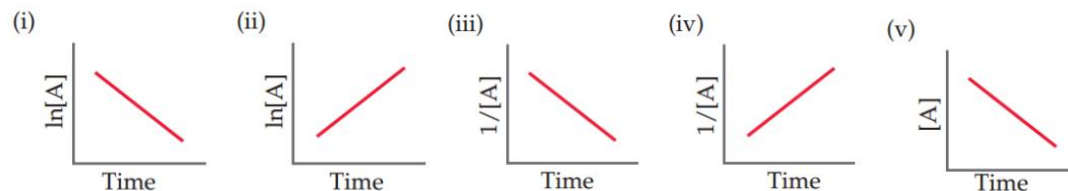


- A) $-1/7\Delta[\text{O}_2]/\Delta t$ B) $-1/4\Delta[\text{NO}_2]/\Delta t$ C) $1/6\Delta[\text{H}_2\text{O}]/\Delta t$ D) $-1/4\Delta[\text{NH}_3]/\Delta t$
 E) All of the above are not valid expressions of the reaction rate.

7) Choose the graph showing the reaction pathway for an overall endothermic reaction with two intermediates that are produced at different rates.



8) Which one of the following graphs shows the correct relationship between concentration and time for a reaction that is first order in [A]?



9) The second order reaction $2 \text{CH}_4 \rightarrow \text{C}_2\text{H}_2 + 3 \text{H}_2$ has a rate constant of $5.76 \text{ M}^{-1}\cdot\text{min}^{-1}$ at 1600K. How long (in min) will it take for the concentration of CH_4 to reduce from 0.89 M to $5.25 \times 10^{-4} \text{ M}$?

- A) 25 B) 67 C) 3.3×10^2 D) 4.2×10^4 E) 3.9

10) The rate of a reaction depends on _____.

- A) collision frequency
 B) collision energy
 C) collision orientation
 D) all of the above
 E) none of the above

11) The rate law of a reaction is $\text{rate} = k[\text{A}]^2[\text{X}]$. The units of the rate constant are _____.

- A) $\text{mol}\cdot\text{L}^{-1}\cdot\text{s}^{-1}$
 B) $\text{L}\cdot\text{mol}^{-1}\cdot\text{s}^{-1}$
 C) $\text{mol}^2\cdot\text{L}^{-2}\cdot\text{s}^{-1}$
 D) $\text{mol}\cdot\text{L}^{-1}\cdot\text{s}^{-2}$
 E) $\text{L}^2\cdot\text{mol}^{-2}\cdot\text{s}^{-1}$

12) Based on their activation energies and energy changes and assuming that all collision factors are the same, which of the following reactions would be fastest in the reverse direction?

- A) $E_a = E_{a1} = 47 \text{ kJ/mol}$, $\Delta E = \Delta E_1 = -25 \text{ kJ/mol}$.
 B) $E_a = E_{a2} = 30 \text{ kJ/mol}$, $\Delta E = \Delta E_2 = -10 \text{ kJ/mol}$.
 C) $E_a = E_{a3} = 53 \text{ kJ/mol}$, $\Delta E = \Delta E_3 = 13 \text{ kJ/mol}$.
 D) $E_{a2} = 30 \text{ kJ/mol}$, $\Delta E = \Delta E_2 = -10 \text{ kJ/mol}$ and $E_{a3} = 53 \text{ kJ/mol}$, $\Delta E = \Delta E_3 = 13 \text{ kJ/mol}$.

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- 13) A catalyst can increase the rate of a reaction _____.
 A) by changing the value of the frequency factor (A)
 B) by increasing the overall activation energy (E_a of the reaction)
 C) by lowering the activation energy of the reverse reaction
 D) by providing an alternative pathway with a lower activation energy
 E) All of these are ways that a catalyst might act to increase the rate of reaction.
- 14) Which one of the following statements is *incorrect*?
 A) The rate does not depend on the magnitude of the ΔE , the internal energy change for the overall reaction.
 B) The lower the activation energy, the faster the rate if equal numbers of orientation independent collisions are always involved.
 C) More molecules in a gas have energies above some threshold value as the temperature is increased.
 D) Activation energies of simple reactions can be negative.
 E) The rates of two reactions can be equal at one temperature but differ at another.
- 15) Using the data, which of the following is the rate constant for the rearrangement of methyl isonitrile at 320 °C ?

T (K)	1/T (K ⁻¹)	ln k
462.9	2.160×10^{-3}	-10.986
472.1	2.118×10^{-3}	-10.178
503.5	1.986×10^{-3}	-7.637
524.4	1.907×10^{-3}	-6.117

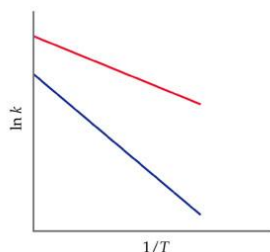
- A) $8.1 \times 10^{-15} \text{ s}^{-1}$ B) $1.5 \times 10^{-13} \text{ s}^{-1}$ C) $2.7 \times 10^{-9} \text{ s}^{-1}$ D) $1.6 \times 10^{-1} \text{ s}^{-1}$ E) $9.2 \times 10^3 \text{ s}^{-1}$

- 16) Observe the following reactants and predict which of the following may form the potential products given the values for the energy of activation, E_a , and enthalpy of the reaction, ΔH , in combination with the molecular orientation.



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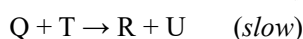
17) As shown in the figure below, _____ reaction has the larger value for E_a and _____ reaction has the larger value for the frequency factor, A



- A) blue, red B) blue, blue C) red, blue D) red, red

18) Consider the following hypothetical reaction: $2 P + Q \rightarrow 2 R + S$

The following mechanism is proposed for this reaction:

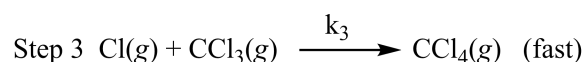
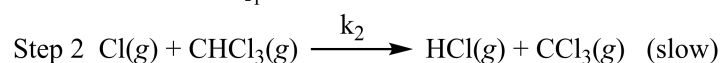
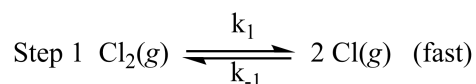


Substances T and U are unstable intermediates. What rate law is predicted by this mechanism?

- A) Rate = $k[P]^2$ B) Rate = $k[P][Q]$ C) Rate = $k[P]^2[Q]$ D) Rate = $k[P][Q]^2$ E) Rate = $k[U]$

Part B short question (20 points)

19) The following mechanism has been proposed for the gas phase reaction of chloroform ($CHCl_3$) and chlorine:



(a) What is the overall reaction? (b) What are the intermediates in the mechanism? (c) What is the molecularity of each of the elementary reactions? (d) What is the rate determining step? (e) What is the rate law predicted by this mechanism?

20) Consider the following hypothetical aqueous reaction: $A(aq) \rightarrow B(aq)$. A flask is charged with 0.065 mol of A in a total volume of 100.0 mL. The following data are collected:

Time (min)	0	10	20	30	40
Moles of A	0.065	0.051	0.042	0.036	0.031

(a) determine whether the reaction is first order or second order. (b) Calculate the rate constant for the reaction with proper unit. (c) What is the half-life for the reaction?

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20) The first order rate constant for the decomposition of N_2O_5 , $2 \text{N}_2\text{O}_5(g) \rightarrow 4 \text{NO}_2(g) + \text{O}_2(g)$, at 70°C is $6.82 \times 10^{-3} \text{ s}^{-1}$. Suppose we start with 0.0250 mol of $\text{N}_2\text{O}_5(g)$ in a volume of 2.00 L. (a) How many moles of N_2O_5 will remain after 5.0 min? (b) What is the half-life of N_2O_5 at 70°C ? (c) How many minutes will it take for the quantity of N_2O_5 to drop to 0.0100 mol?

21) The gas-phase decomposition of ozone is thought to occur by the following two-step mechanism.

Step 1: $\text{O}_3(g) \rightleftharpoons \text{O}_2(g) + \text{O}(g)$ (fast)

Step 2: $\text{O}(g) + \text{O}_3(g) \rightarrow 2 \text{O}_2(g)$ (slow)

(a) Write the balanced equation for the overall reaction. (*Express your answer as a chemical equation including phases.*) (b) Derive the rate law that is consistent with this mechanism. (c) Is O a catalyst or an intermediate? (d) If instead the reaction occurred in a single step, would the rate law change? If so, what would it be?

Chapter 11 Homework Answer Sheet

Name:_____ Student ID:_____ Instructor:_____ Score:_____

Part A multiple choice (18 points)

1-5_____

6-10_____

11-15_____

16-18_____

Part B short question (20 points)

19 (8 points)

20 (4 points)

21 (3 points)

22 (5 points)